Relations: Questions and Full Solutions

Question 01

English

Let A and B be finite sets such that $n(A) = 5$, $n(B) = 7$ and $n\big((A \times B) \cup (B \times A)\big) = 54$, where $n(X)$ denotes the number of elements of a finite set X. Then $n\big((A \cup B) \times (A \cap B)\big)$ is equal to

(A) 24
(B) 28
(C) 32
(D) 36

Only one option is correct.

हिन्दी

माना A तथा B ऐसे परिमित समुच्चय हैं कि $n(A) = 5$, $n(B) = 7$ तथा $n\big((A \times B) \cup (B \times A)\big) = 54$, जहाँ $n(X)$ परिमित समुच्चय X के अवयवों की संख्या को दर्शाता है। तब $n\big((A \cup B) \times (A \cap B)\big)$ बराबर है

(A) 24
(B) 28
(C) 32
(D) 36

केवल एक विकल्प सही है।

Solution

For finite sets $n(X \times Y) = n(X)\, n(Y)$, so $n(A \times B) = 5 \cdot 7 = 35$ and $n(B \times A) = 7 \cdot 5 = 35$.

An ordered pair (x, y) lies in $A \times B$ and in $B \times A$ exactly when $x \in A$, $y \in B$, $x \in B$ and $y \in A$, that is, when $x \in A \cap B$ and $y \in A \cap B$. Hence $(A \times B) \cap (B \times A) = (A \cap B) \times (A \cap B)$.

Let $k = n(A \cap B)$. Then $n\big((A \times B) \cap (B \times A)\big) = k^2$ and, by the formula for the number of elements of a union,

$n\big((A \times B) \cup (B \times A)\big) = 35 + 35 - k^2 = 70 - k^2$.

$70 - k^2 = 54 \Rightarrow k^2 = 16 \Rightarrow k = 4$ (and $k = 4 \leq n(A) = 5$ is possible).

$n(A \cup B) = n(A) + n(B) - n(A \cap B) = 5 + 7 - 4 = 8$.

$n\big((A \cup B) \times (A \cap B)\big) = 8 \cdot 4 = 32$. (The value 28 comes from using $n(B)$ in place of $n(A \cup B)$, and 36 from taking $n(A \cup B) = 9$.)

Correct Option: (C)

Question 02

English

Let $A = \{1, 2, 3\}$. Let N denote the number of subsets S of $A \times A$ such that $(1, 2) \notin S$ and $S = X \times Y$ for some subsets X and Y of A. Then the value of N is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $A = \{1, 2, 3\}$ है। माना N, $A \times A$ के उन उपसमुच्चयों S की संख्या है जिनके लिए $(1, 2) \notin S$ है तथा A के किन्हीं उपसमुच्चयों X तथा Y हेतु $S = X \times Y$ है। तब N का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

A subset of $A \times A$ of the form $X \times Y$ may arise from several pairs (X, Y), so the pairs (X, Y) cannot simply be counted.

Since $n(X \times Y) = n(X)\, n(Y)$, the product is empty exactly when $X = \emptyset$ or $Y = \emptyset$. All such pairs (X, Y) (there are $2^3 + 2^3 - 1 = 15$ of them) produce the single subset $\emptyset$.

If $X \neq \emptyset$ and $Y \neq \emptyset$, then X is the set of first entries of the pairs in $X \times Y$ and Y is the set of second entries. So $X \times Y$ determines X and Y, and distinct pairs of non-empty subsets give distinct products.

A has $2^3 = 8$ subsets, $2^3 - 1 = 7$ of them non-empty, so there are $7 \cdot 7 = 49$ non-empty products. Together with $\emptyset$: 50 subsets of $A \times A$ are of the form $X \times Y$.

Among them, $(1, 2) \in X \times Y$ iff $1 \in X$ and $2 \in Y$ (then both are non-empty). Subsets of A containing 1: $2^2 = 4$; containing 2: $2^2 = 4$. So 16 products contain $(1, 2)$. The empty set does not contain $(1, 2)$.

$N = 50 - 16 = 34$.

Final Answer: 34

Question 03

English

Let a and b be real numbers and let R be the relation on $\mathbb{R}$ given by $R = \{(x, y) \in \mathbb{R} \times \mathbb{R} : (x^2 + ax + b)\, y = 4x - 3\}$. Suppose that the domain of R is $\mathbb{R} \setminus \{-1\}$ and that R is not a function from its domain to $\mathbb{R}$. Then the value of $a + b$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना a तथा b वास्तविक संख्याएँ हैं तथा $\mathbb{R}$ पर संबंध $R = \{(x, y) \in \mathbb{R} \times \mathbb{R} : (x^2 + ax + b)\, y = 4x - 3\}$ है। माना कि R का प्रांत $\mathbb{R} \setminus \{-1\}$ है तथा R अपने प्रांत से $\mathbb{R}$ में फलन नहीं है। तब $a + b$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Write $q(x) = x^2 + ax + b$. For a fixed real x_0, the elements of R with first entry x_0 are the solutions y of $q(x_0)\, y = 4x_0 - 3$.

Case $q(x_0) \neq 0$: exactly one solution $y = \dfrac{4x_0 - 3}{q(x_0)}$, so x_0 is in the domain with one image.

Case $q(x_0) = 0$ and $x_0 \neq \frac{3}{4}$: the equation reads $0 = 4x_0 - 3 \neq 0$, no solution, so x_0 is not in the domain.

Case $q(x_0) = 0$ and $x_0 = \frac{3}{4}$: the equation reads $0 = 0$, true for every y; the whole vertical line $x = \frac{3}{4}$ lies in R, so $\frac{3}{4}$ is in the domain with infinitely many images.

So the domain of R is $\mathbb{R}$ minus the roots of q different from $\frac{3}{4}$, and R fails to be a function exactly when $\frac{3}{4}$ is a root of q.

Domain $= \mathbb{R} \setminus \{-1\}$: -1 must be a root of q, and the other root (if different) cannot be removed from the domain, so it must be $\frac{3}{4}$. Hence $q(x) = (x + 1)^2$ or $q(x) = (x + 1)\left(x - \frac{3}{4}\right)$.

If $q(x) = (x + 1)^2$, then $\frac{3}{4}$ is not a root, every $x \neq -1$ has exactly one image and R is the function $y = \dfrac{4x - 3}{(x + 1)^2}$; this contradicts the hypothesis.

Hence $q(x) = (x + 1)\left(x - \frac{3}{4}\right) = x^2 + \frac{1}{4}x - \frac{3}{4}$, where the vertical line $x = \frac{3}{4}$ meets R in infinitely many points, so R is not a function.

$a = \frac{1}{4}$, $b = -\frac{3}{4}$, $a + b = -\frac{1}{2} = -0.5$.

Final Answer: -0.5

Question 04

English

Let $A = \{1, 2, 3, 4\}$ and let S be the set of all relations on A that are both symmetric and transitive. Let p be the number of relations in S that are antisymmetric, and let q be the number of relations in S that are not reflexive. Then the value of $p - q$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $A = \{1, 2, 3, 4\}$ है तथा S, A पर उन सभी संबंधों का समुच्चय है जो सममित तथा संक्रामक दोनों हैं। माना p, S के उन संबंधों की संख्या है जो प्रतिसममित हैं, तथा q, S के उन संबंधों की संख्या है जो स्वतुल्य नहीं हैं। तब $p - q$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $R \in S$. If $(a, b) \in R$, symmetry gives $(b, a) \in R$, and transitivity applied to $(a, b), (b, a)$ gives $(a, a) \in R$; similarly $(b, b) \in R$.

Let B be the set of elements of A that occur in some pair of R. By the previous line R is reflexive on B, and it is symmetric and transitive, so R is an equivalence relation on B (and contains no pair with an entry outside B).

Conversely, for any $B \subseteq A$ and any equivalence relation on B, the same set of pairs is a symmetric and transitive relation on A. Different B or different relations give different members of S.

An equivalence relation on a set is the same as a partition of the set into equivalence classes. Number of partitions of a k-element set: $k = 0$: 1; $k = 1$: 1; $k = 2$: 2; $k = 3$: 5 ($\{abc\}$, three of type $\{ab\}\{c\}$, $\{a\}\{b\}\{c\}$); $k = 4$: 15 (1 of type 4, 4 of type $3 + 1$, 3 of type $2 + 2$, 6 of type $2 + 1 + 1$, 1 of type $1 + 1 + 1 + 1$).

$n(S) = \binom{4}{0} \cdot 1 + \binom{4}{1} \cdot 1 + \binom{4}{2} \cdot 2 + \binom{4}{3} \cdot 5 + \binom{4}{4} \cdot 15 = 1 + 4 + 12 + 20 + 15 = 52$.

$R \in S$ is reflexive iff $B = A$, i.e. R is an equivalence relation on A: 15 relations. Hence $q = 52 - 15 = 37$.

If $R \in S$ is antisymmetric, a pair (a, b) with $a \neq b$ would force $(b, a) \in R$ by symmetry, contradicting antisymmetry. So R consists of diagonal pairs only. Conversely every subset of $\{(1, 1), (2, 2), (3, 3), (4, 4)\}$ is symmetric, antisymmetric and transitive. Hence $p = 2^4 = 16$.

$p - q = 16 - 37 = -21$.

Final Answer: -21

Question 05

English

Let $A = \{1, 2, 3\}$. For a relation R on A, let $m(R)$ denote the least number of ordered pairs from $A \times A$ that must be added to R so that the resulting relation is an equivalence relation on A. Then the arithmetic mean of $m(R)$, taken over all symmetric relations R on A, is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $A = \{1, 2, 3\}$ है। A पर किसी संबंध R के लिए माना $m(R)$, $A \times A$ के उन क्रमित युग्मों की न्यूनतम संख्या है जिन्हें R में जोड़ने पर प्राप्त संबंध A पर एक तुल्यता संबंध हो जाए। तब A पर सभी सममित संबंधों R पर लिया गया $m(R)$ का समांतर माध्य _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

A symmetric relation on A is determined by which of the 3 diagonal pairs it contains and by which of the 3 unordered pairs $\{1, 2\}, \{1, 3\}, \{2, 3\}$ it contains (each contributing both ordered pairs). So there are $2^3 \cdot 2^3 = 64$ symmetric relations.

Any equivalence relation containing R contains the diagonal and, by transitivity, all pairs of elements that are linked by a chain of pairs of R. The smallest one E_R therefore has as classes the groups of elements linked through the off-diagonal pairs of R, and $n(E_R)$ is the sum of the squares of the class sizes. Hence $m(R) = n(E_R) - n(R)$.

$n(E_R)$ depends only on the set E of unordered pairs: $E = \emptyset$ gives classes $1, 1, 1$ and $n(E_R) = 3$; one unordered pair gives classes $2, 1$ and $n(E_R) = 5$ (3 choices of E); two or three unordered pairs link all three elements, $n(E_R) = 9$ (4 choices of E).

Mean of $n(E_R)$ over the 64 relations (each E occurs with 8 diagonal parts): $\dfrac{1 \cdot 3 + 3 \cdot 5 + 4 \cdot 9}{8} = \dfrac{54}{8} = 6.75$.

Mean of $n(R)$: each of the 3 diagonal pairs lies in exactly half of the relations, and each of the 3 unordered pairs lies in half of them and contributes 2 ordered pairs. Mean $= 3 \cdot \frac{1}{2} + 3 \cdot 2 \cdot \frac{1}{2} = 4.5$.

Mean of $m(R) = 6.75 - 4.5 = 2.25$. (Check: for $R = \emptyset$, $m = 3$; for $R = A \times A$, $m = 0$.)

Final Answer: 2.25

Question 06

English

Let $A = \{1, 2, 3, 4, 5, 6\}$. Let p be the number of equivalence relations on A having exactly three equivalence classes, each of which contains at least two elements, and let q be the number of equivalence relations on A having exactly two equivalence classes, each of which contains at least two elements. Then the value of $\frac{p}{q}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $A = \{1, 2, 3, 4, 5, 6\}$ है। माना p, A पर ऐसे तुल्यता संबंधों की संख्या है जिनमें ठीक तीन तुल्यता वर्ग हैं तथा प्रत्येक वर्ग में कम से कम दो अवयव हैं, तथा q, A पर ऐसे तुल्यता संबंधों की संख्या है जिनमें ठीक दो तुल्यता वर्ग हैं तथा प्रत्येक वर्ग में कम से कम दो अवयव हैं। तब $\frac{p}{q}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Every equivalence relation on A determines the partition of A into its equivalence classes, and every partition determines exactly one equivalence relation. So p and q count partitions.

Three classes, each with at least 2 elements, use at least 6 elements; as $n(A) = 6$, every class has exactly 2 elements. Choose the classes: $\binom{6}{2}\binom{4}{2}\binom{2}{2} = 90$ ordered choices, and the 3! orders of the same three classes give the same partition: $p = \dfrac{90}{6} = 15$.

Two classes: the class C that does not contain 6 determines the partition (the other class is $A \setminus C$). C can be any non-empty subset of $\{1, 2, 3, 4, 5\}$, so there are $2^5 - 1 = 31$ partitions into two classes.

A class with one element occurs when C is a singleton (5 ways) or when the class of 6 is $\{6\}$, i.e. $C = \{1, 2, 3, 4, 5\}$ (1 way). These 6 cases are excluded: $q = 31 - 6 = 25$. (Check: $\binom{6}{2} + \frac{1}{2}\binom{6}{3} = 15 + 10 = 25$ for the types $2 + 4$ and $3 + 3$.)

$\dfrac{p}{q} = \dfrac{15}{25} = \dfrac{3}{5} = 0.6.$

Final Answer: 0.6

Question 07

English

Let $A = \{1, 2, 3, \ldots, 40\}$ and let R be the relation on A defined by $(a, b) \in R$ if and only if b is the largest odd divisor of a. Let R^{-1} denote the inverse of R, and let S be the relation on A consisting of all ordered pairs $(a, c) \in A \times A$ for which there exists $b \in A$ with $(a, b) \in R$ and $(b, c) \in R^{-1}$. Then the number of elements in S is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $A = \{1, 2, 3, \ldots, 40\}$ है तथा A पर संबंध R इस प्रकार परिभाषित है कि $(a, b) \in R$ यदि और केवल यदि b, a का सबसे बड़ा विषम भाजक है। माना R^{-1}, R का प्रतिलोम संबंध है तथा S, A पर वह संबंध है जिसमें वे सभी क्रमित युग्म $(a, c) \in A \times A$ हैं जिनके लिए कोई $b \in A$ ऐसा है कि $(a, b) \in R$ तथा $(b, c) \in R^{-1}$ है। तब S में अवयवों की संख्या _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For $a \in A$ write $a = 2^r m$ with m odd; then $m = o(a)$ is the largest odd divisor of a and $o(a) \le a \le 40$, so $o(a) \in A$. Thus $(a, b) \in R \iff b = o(a)$, and each a has exactly one image.

By definition of the inverse, $(b, c) \in R^{-1} \iff (c, b) \in R \iff b = o(c)$.

So $(a, c) \in S$ iff some $b \in A$ satisfies $b = o(a)$ and $b = o(c)$, i.e. iff $o(a) = o(c)$.

"Having the same largest odd divisor" is reflexive, symmetric and transitive, so S is an equivalence relation on A. The class of an odd m is $\{m, 2m, 4m, \ldots\} \cap A$, and a class with s elements contributes s^2 ordered pairs.

$m = 1$: $\{1, 2, 4, 8, 16, 32\}$, $s = 6$. $m = 3$: $\{3, 6, 12, 24\}$, $m = 5$: $\{5, 10, 20, 40\}$, $s = 4$. $m = 7$: $\{7, 14, 28\}$, $m = 9$: $\{9, 18, 36\}$, $s = 3$. $m = 11, 13, 15, 17, 19$: $\{m, 2m\}$, $s = 2$. $m = 21, 23, \ldots, 39$: $\{m\}$, $s = 1$, ten classes. (Total $6 + 8 + 6 + 10 + 10 = 40$ elements.)

$n(S) = 6^2 + 2 \cdot 4^2 + 2 \cdot 3^2 + 5 \cdot 2^2 + 10 \cdot 1^2 = 36 + 32 + 18 + 20 + 10 = 116$.

Final Answer: 116

Question 08

English

For each integer k, let R_k be the relation on $\mathbb{Z}$ defined by $R_k = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : x^2 + y^2 \leq 2xy + k\}$. Then the sum of all $k \in \{-5, -4, \ldots, 4, 5\}$ for which R_k is transitive is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

प्रत्येक पूर्णांक k के लिए माना $\mathbb{Z}$ पर संबंध $R_k = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : x^2 + y^2 \leq 2xy + k\}$ द्वारा परिभाषित है। तब उन सभी $k \in \{-5, -4, \ldots, 4, 5\}$ का योग, जिनके लिए R_k संक्रामक है, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$x^2 + y^2 - 2xy = (x - y)^2$, so $(x, y) \in R_k \iff (x - y)^2 \leq k$.

If $k < 0$, no integers satisfy $(x - y)^2 \leq k$, so R_k is the empty relation. A relation is transitive when $(x, y), (y, z) \in R$ always forces $(x, z) \in R$; for the empty relation there is no such pair, so the condition holds. Hence $k = -5, -4, -3, -2, -1$ all qualify.

If $k = 0$, $(x - y)^2 \leq 0$ means $x = y$, so $R_0 = \{(x, x) : x \in \mathbb{Z}\}$ is the identity relation, which is transitive.

If $k \in \{1, 2, 3\}$: $(0, 1)$ and $(1, 2)$ lie in R_k (difference 1, square $1 \leq k$) but $(0, 2)$ has square $4 > k$, so R_k is not transitive.

If $k \in \{4, 5\}$: $(0, 2)$ and $(2, 4)$ lie in R_k (square $4 \leq k$) but $(0, 4)$ has square $16 > k$, so R_k is not transitive.

Required sum $= (-5) + (-4) + (-3) + (-2) + (-1) + 0 = -15$. (Omitting the empty relations would give 0.)

Final Answer: -15

Question 09

English

Let $A = \{1, 2, 3, 4\}$ and let $I_A = \{(a, a) : a \in A\}$ denote the identity relation on A. For a relation R on A, let R^{-1} denote its inverse. Then the number of relations R on A satisfying $R \cup R^{-1} = A \times A$ and $R \cap R^{-1} = I_A$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $A = \{1, 2, 3, 4\}$ है तथा $I_A = \{(a, a) : a \in A\}$, A पर तत्समक संबंध है। A पर किसी संबंध R के लिए माना R^{-1} उसका प्रतिलोम संबंध है। तब A पर ऐसे संबंधों R की संख्या, जो $R \cup R^{-1} = A \times A$ तथा $R \cap R^{-1} = I_A$ को संतुष्ट करते हैं, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$(x, y) \in R^{-1} \iff (y, x) \in R$. In particular $(a, a) \in R^{-1} \iff (a, a) \in R$.

Diagonal: $(a, a) \in R \cap R^{-1} \iff (a, a) \in R$. Since $R \cap R^{-1} = I_A$ contains every (a, a), R contains every (a, a), i.e. R is reflexive; the union condition on the diagonal is then also satisfied.

Off the diagonal, take $a \neq b$. $(a, b) \in R \cup R^{-1}$ iff $(a, b) \in R$ or $(b, a) \in R$; $(a, b) \in R \cap R^{-1}$ iff both are in R.

$R \cup R^{-1} = A \times A$ requires at least one of $(a, b), (b, a)$ in R, and $R \cap R^{-1} = I_A$ (which has no off-diagonal pair) forbids both. So exactly one of them is in R (the relation is reflexive, antisymmetric and any two distinct elements are comparable).

The choices for the $\binom{4}{2} = 6$ unordered pairs are independent, 2 ways each: $2^6 = 64$ relations.

Final Answer: 64